

An Introduction to Differential Geometry and Collar Neighbourhoods

Hannah Scholz

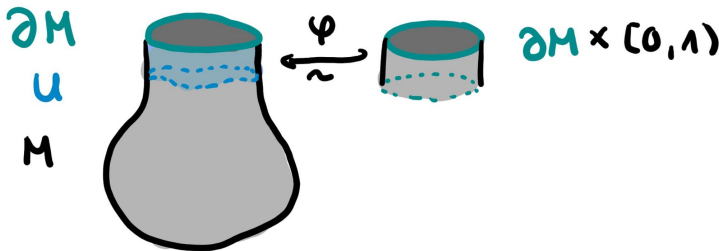
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Collar Neighbourhood

Definition: Collar Neighbourhood

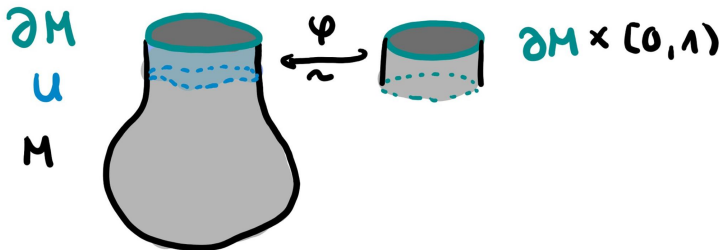
Let M be a smooth manifold with boundary. Let U be a neighbourhood of the boundary ∂M . U is called a *collar neighbourhood* if there is a diffeomorphism $\varphi: \partial M \times [0, 1) \rightarrow U$ such that $\varphi(p, 0) = p$ for all $p \in \partial M$.



Collar Neighbourhood

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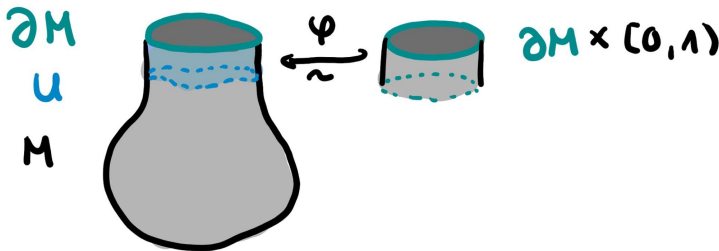
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Collar Neighbourhood Theorem

Collar Neighbourhood Theorem

Let M be a smooth manifold with non-empty boundary. Then ∂M has a collar neighbourhood.

Topological Manifold with Boundary

Definition: Topological Manifold with Boundary

A topological space M is a *topological n -manifold* if

- (i) M is Hausdorff.
- (ii) M is second-countable, i.e. M has a countable base.
- (iii) M is locally Euclidean, i.e. for every point p there is a neighbourhood of p that is homeomorphic to an open set in $\mathbb{H}^n$.

Definition: Chart

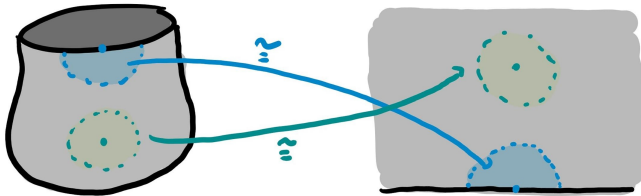
Let M be a topological manifold with boundary. Let U be an open subset of M and $\varphi: U \rightarrow \mathbb{H}^n$ a homeomorphism onto an open subset of $\mathbb{H}^n$. Then we call (U, φ) a chart of M .

Boundary of a Topological Manifold

Definition: Boundary

Let M be a topological manifold with boundary.

- (i) A point $p \in M$ is a *boundary point* if it is in the domain of a *boundary chart*, i.e. a chart (U, φ) with $\varphi(p) \in \partial \mathbb{H}^n$.
- (ii) A point $p \in M$ is an *interior point* if it is in the domain of an *interior chart*, i.e. a chart (U, φ) with $\varphi(U) \cap \partial \mathbb{H}^n = \emptyset$.



Smooth Manifold

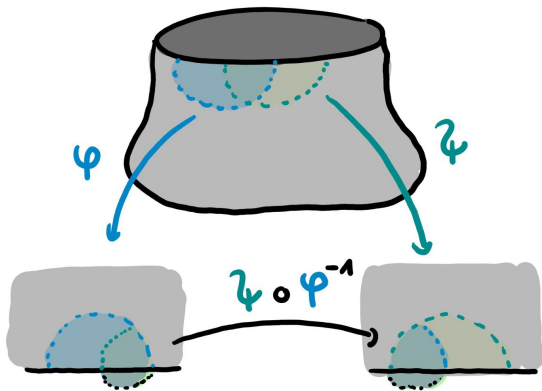
Definition: Smooth Atlas

Let M be a topological manifold. Let $\mathcal{A}$ be a collection of charts whose domains cover M . $\mathcal{A}$ is a smooth atlas for M if any two charts (φ, U) and (ψ, V) are *smoothly compatible*, i.e. either $U \cap V$ is empty or $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is smooth in the ordinary sense.

Definition: Smooth Manifold

A pair $(M, \mathcal{A})$ of a topological manifold M and a smooth atlas $\mathcal{A}$ on M , which is maximal with respect to inclusion, is called a *smooth manifold*.

Compatible Charts



Compatibility of charts

Smooth Maps

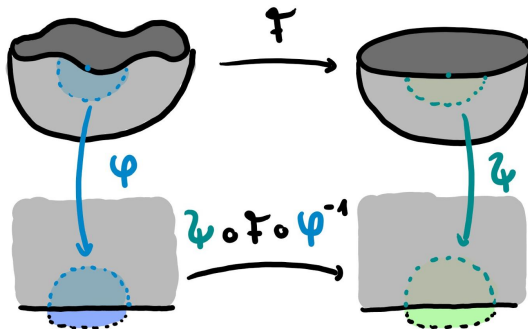
Definition: Smooth Map

Let $(M, \mathcal{A}_M)$ and $(N, \mathcal{A}_N)$ be smooth manifolds. A map $F: M \rightarrow N$ is called *smooth* if for each $p \in M$ there are charts $(U, \varphi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(p) \in V$ and $\psi \circ F \circ \varphi^{-1}: \varphi(U) \rightarrow \psi(V)$ is smooth.

Definition: Diffeomorphism

A *diffeomorphism* between two smooth manifolds $(M, \mathcal{A}_M)$ and $(N, \mathcal{A}_N)$ is a smooth bijective map that has a smooth inverse.

Smooth Maps



Local Smoothness

Collar Neighbourhoods

Definition: Collar Neighbourhood

Let M be a smooth manifold with boundary. Let U be a neighbourhood of the boundary ∂M . U is called a *collar neighbourhood* if there is a diffeomorphism $\varphi: \partial M \times [0, 1) \rightarrow U$ such that $\varphi(p, 0) = p$ for all $p \in \partial M$.

Collar Neighbourhood Theorem

Let M be a smooth manifold with non-empty boundary. Then ∂M has a collar neighbourhood.

Cobordism

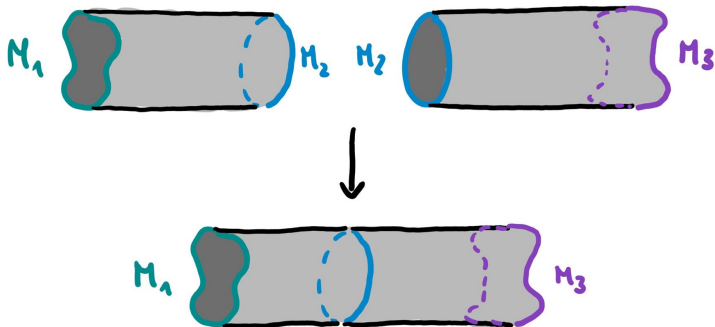
Definition: Cobordism

Smooth manifolds M and N are called *cobordant* if there is a compact smooth manifold W such that ∂W is diffeomorphic to $M \sqcup N$.

Lemma

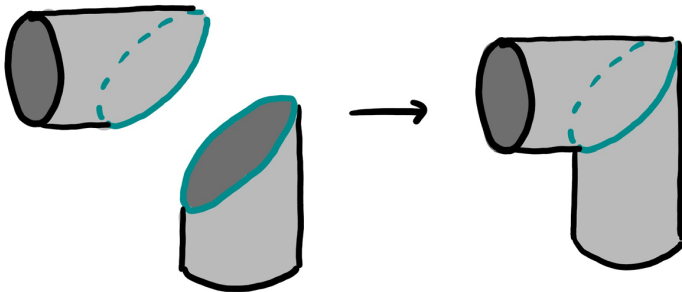
Cobordism is an equivalence relation.

Transitivity of Cobordism



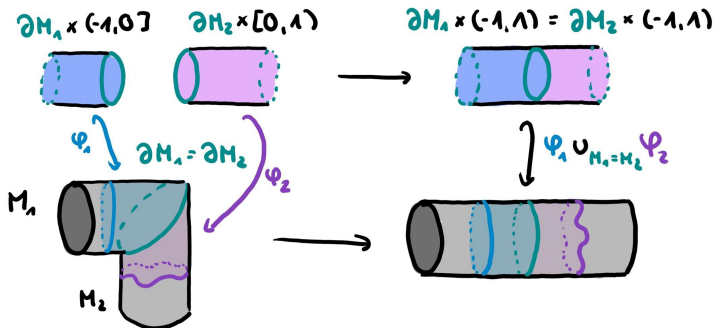
Transitivity of Cobordism

Glueing along Boundary: The Problem



Glueing topologically doesn't give smoothness

Glueing along Boundary: The Solution



Collar neighbourhoods allow us to define a smooth structure

Tangent Bundles

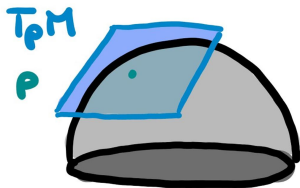
Definition: Derivation

A *derivation* at a point p in a smooth manifold with boundary is a linear map $v: C^\infty(M) \rightarrow \mathbb{R}$ if for all $f, g \in C^\infty(M)$ we have $v(fg) = f(p)v(g) + v(f)g(p)$.

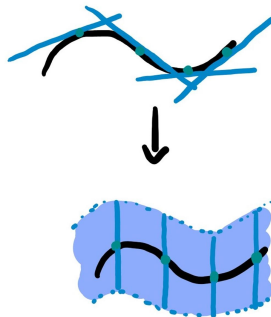
Definition: Tangent Space

- (i) We denote the set of all derivations at $p \in M$ as $T_p M$ and call it the *tangent space* to M at p .
- (ii) The *tangent bundle* of M is the disjoint union of the tangent spaces at all points of M : $TM := \coprod_{p \in M} T_p M$.

Tangent Bundle



Tangent space $T_p M$ at p

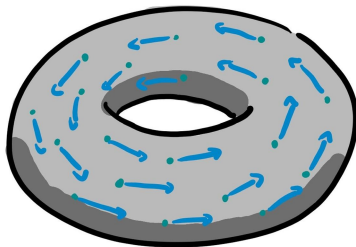


Tangent bundle TM

Vector field

Definition: Smooth vector field

Let M be a smooth manifold. Then a *smooth vector field* on M is a smooth map $X: M \rightarrow TM$ such that for every $p \in M$, $X(p) \in T_pM$.



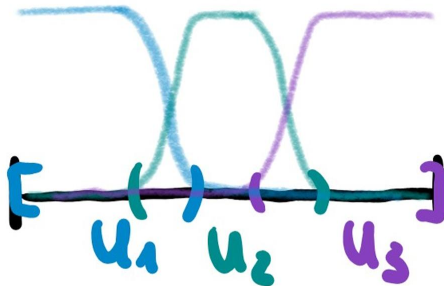
Partition of Unity

Definition: Partition of Unity

Let M be a topological space and $\mathcal{X}$ a cover of M . Then a *partition of unity subordinate to $\mathcal{X}$* is a family of functions $(\phi_X: M \rightarrow \mathbb{R})_{X \in \mathcal{X}}$ such that

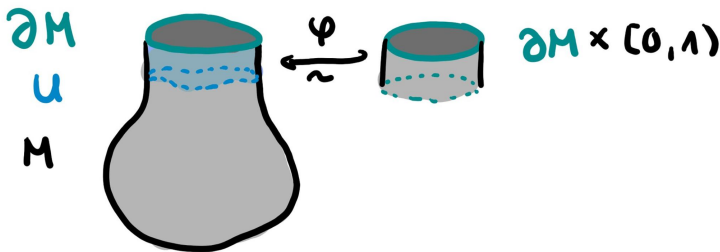
- (i) $\phi_X(p) \in [0, 1]$ for all $p \in M$ and $X \in \mathcal{X}$
- (ii) $\overline{\{p \in M \mid \phi_X(p) \neq 0\}} \subseteq X$ for all $X \in \mathcal{X}$
- (iii) For every $p \in M$ there is a neighbourhood that intersects only finitely many of $\overline{\{p \in M \mid \phi_X(p) \neq 0\}}$.
- (iv) $\sum_{X \in \mathcal{X}} \phi_X(p) = 1$ for all $p \in M$.

Partition of Unity



A partition of unity

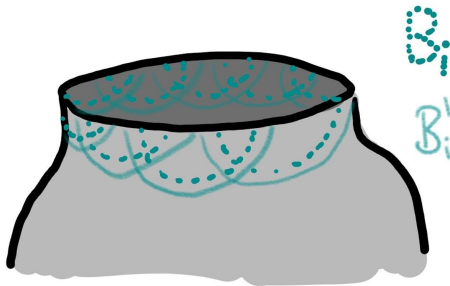
Collar Neighbourhoods



A collar neighbourhood

Proof of the Collar Neighbourhood Theorem

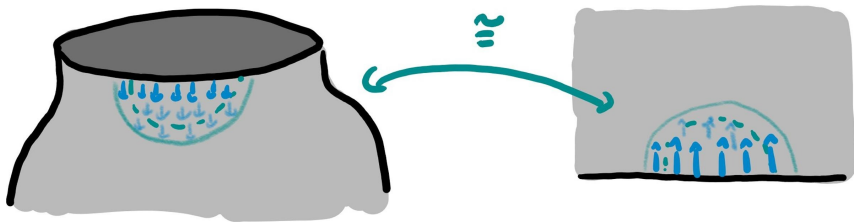
First Step: Define a vector field that is inward-pointing on the boundary.



Cover the boundary in regular coordinate half-balls

Proof of the Collar Neighbourhood Theorem

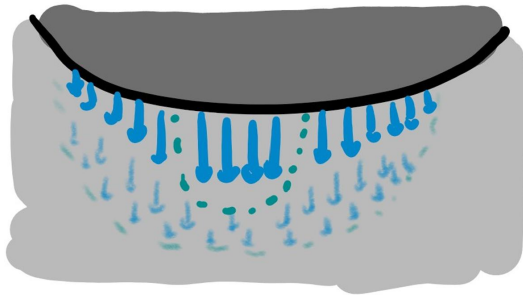
First Step: Define a vector field that is inward-pointing on the boundary.



Find a vector field locally

Proof of the Collar Neighbourhood Theorem

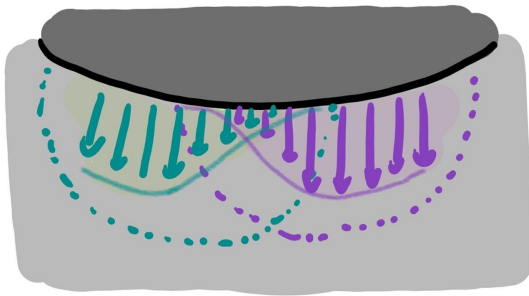
First Step: Define a vector field that is inward-pointing on the boundary.



Multiply the local vector field with a bump function to extend it into a global vector field

Proof of the Collar Neighbourhood Theorem

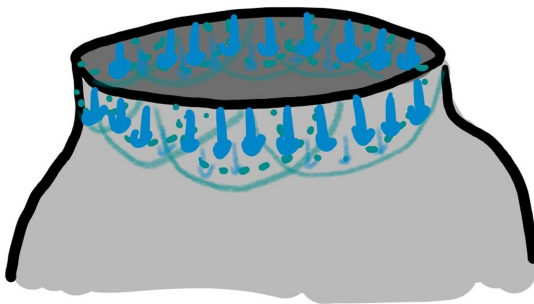
First Step: Define a vector field that is inward-pointing on the boundary.



Multiply the local vector fields with a partition of unity subordinate to $\{B_i\}_i \cup M \setminus \partial M$

Proof of the Collar Neighbourhood Theorem

First Step: Define a vector field that is inward-pointing on the boundary.



Glue together for the desired vector field

Proof of the Collar Neighbourhood Theorem

Second Step: Apply the Boundary Flowout Theorem.

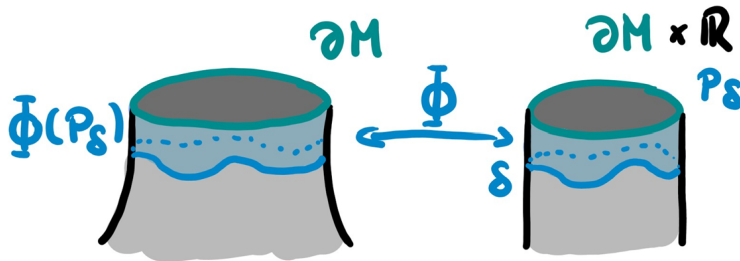
Boundary Flowout Theorem

Let M be a smooth manifold with non-empty boundary. Let X be a smooth vector field on M that is inward-pointing on the boundary. Then we obtain:

- (i) A smooth function $\delta: \partial M \rightarrow \mathbb{R}^+$
- (ii) A smooth embedding $\Phi: \mathcal{P}_\delta \rightarrow M$ where
$$\mathcal{P}_\delta := \{(t, p) \in \mathbb{R} \times \partial M \mid 0 \leq t < \delta(p)\}$$
such that $\Phi(\mathcal{P}_\delta)$ is a neighbourhood of ∂M .

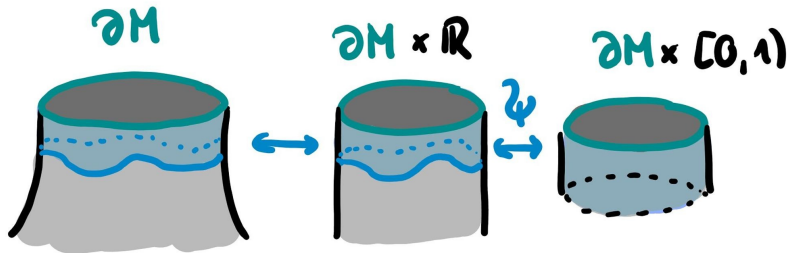
Proof of the Collar Neighbourhood Theorem

Second Step: Apply the Boundary Flowout Theorem.



Proof of the Collar Neighbourhood Theorem

Third Step: Reparametrize $\mathcal{P}_\delta$.



Set $\psi: \partial M \times [0, 1) \rightarrow \mathcal{P}_\delta, (p, t) \mapsto (p, t\delta(t))$.